

How Much Local Data Is *Enough*?

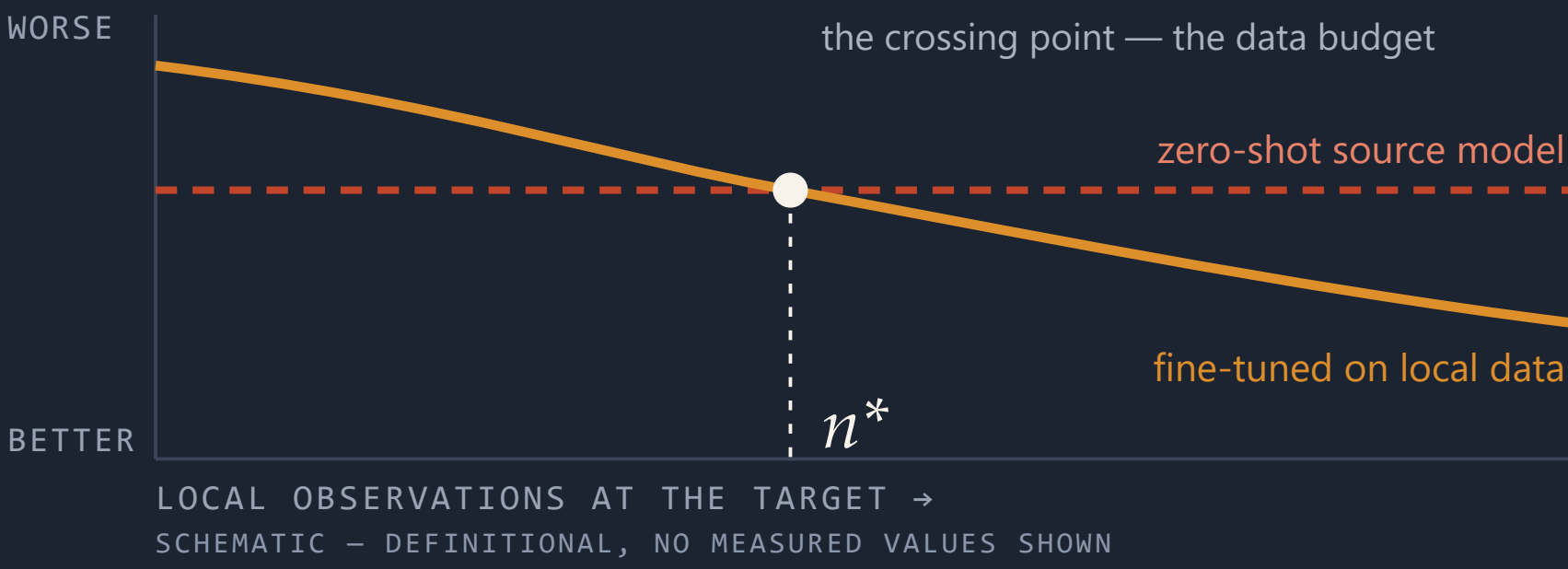
An ontology-grounded neurosymbolic research agenda for transferability in forecasting

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THE QUANTITY — DATA BUDGET

Given a source model and a target site, how much local data must be collected before *fine-tuning beats deploying the source model as-is*?

Call that quantity the site’s **data budget, n^*** : the smallest amount of local data — horizon, altitude range, cloud-regime diversity, seasonal coverage — after which the fine-tuned model consistently beats the zero-shot model.



20 Brazilian sites where the reference budget n^* is already measured, under one fixed protocol
LARGELY COMPLETE

5 rival criteria for what makes two sites **the same kind of place**
TO BE ADJUDICATED AGAINST THOSE BUDGETS

1 symbolic **semantic gate** between a trained model and a new place
PROPOSED · STILL UNDER DEVELOPMENT

01 — THE SETTING

Brazil measures the sun in a few places, and almost nowhere else.

Renewable-energy planning needs reliable solar-irradiation forecasts, and good forecasts need data. A handful of well-instrumented sites have years of clean records; **vast regions have almost nothing** — a continent-sized country with a sparse and uneven measurement network.

02 — THE PRAGMATIC HOPE

Train where data is plentiful, reuse where it is not.

That is **transfer learning** [1]. But reuse is not epistemically neutral: it presupposes that the target is, in some relevant respect, **the same kind of place** as the source.

A theory of n^* would let a planner estimate, **before investing in instrumentation**, how much data a new site will need.

03 — THE HIDDEN CLAIM

“The same kind of place” is not a statistical primitive.

It is a judgment about which properties of a site are **essential** to its solar behavior and which are **incidental** — a question formal ontology was built to clarify [3].

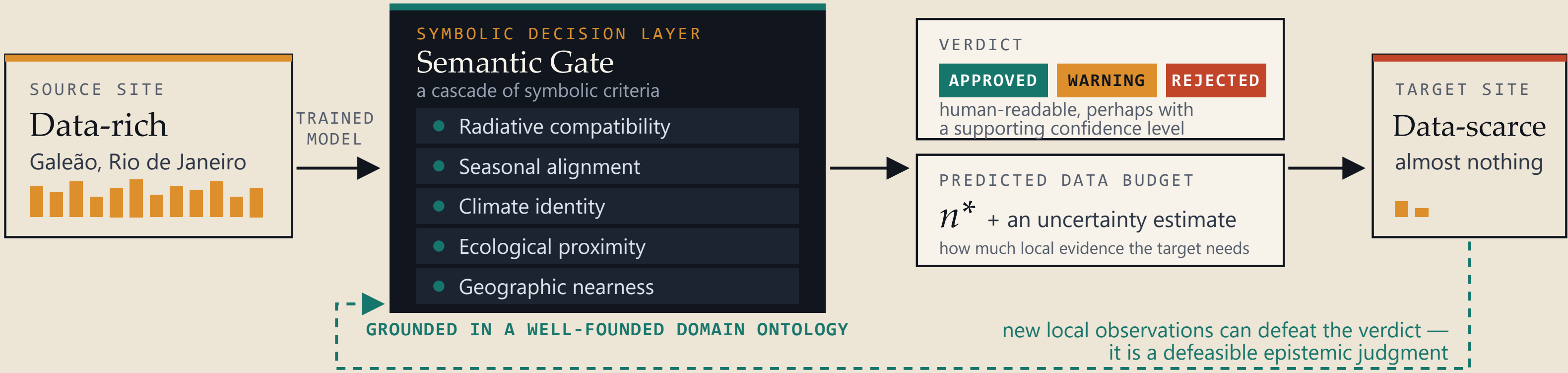
- share a latitude but differ in cloud regime
- share a biome but differ in altitude
- sit far apart yet behave identically

When sites diverge along what matters, transfer **degrades performance rather than improving it** — negative transfer [1, 2].

04 — THE CORE IDEA

A *Neurosymbolic Semantic Gate* between the model and the place.

A symbolic decision layer, still under development, between a trained source model and any prospective target site, reasoning transparently — in terms a domain scientist can read — about whether and how readily transfer is warranted. I propose to ground its symbolic layer in a **well-founded domain ontology** [4], not an ad-hoc feature set, so its commitments are explicit, inspectable and independently auditable [5].



05 — FIVE RIVAL NOTIONS OF SAMENESS

Each criterion is a distinct, *testable hypothesis* about what licenses transfer.

CRITERION	NOTION OF “SAMENESS”	ONTOLOGICAL SOURCE
Radiative compatibility	Comparable top-of-atmosphere forcing	Astronomy / site geometry
Seasonal alignment	Matching shape of the annual cycle	Climatology
Climate identity	Same Köppen class	Köppen–Geiger [6, 7]
Ecological proximity	Near in biome / climate space	Whittaker biomes [8]
Geographic nearness	Small great-circle distance	Spatial proximity

These are **distinct relations that often disagree**. The design keeps them separate so the study can ask which notion of sameness, if any, actually carries predictive weight.

06 — UNDER ADVERSARIAL TEST

Two claims to be put *under test*, not assumed.

H1 – ORDERING

If ontological similarity (as operationalized by the gate) orders source–target pairs, then sites judged **more similar require fewer local observations** to adapt — a measurably lower data budget n^* .

H2 – DISCRIMINATION

If the candidate notions of sameness are **not interchangeable**, some (categorical climate identity) will carry systematically different inductive value from others (graded behavioral proximity) — and the empirical record will differentiate them.

A standard of acceptance that has never been challenged is **indistinguishable from an assumption**.

07 — ADJUDICATION

Will be measured against a *ground truth* that already exists.

The reference budget n^* is established across **20 Brazilian sites** under a fixed protocol, using reanalysis weather data [9]. Each criterion, and the aggregate verdict, will be assessed for its ability to **rank target sites by n^*** — controlling for target-site difficulty, so an apparent effect of similarity is not an artifact of a site being easier to forecast.



- 1 source site · Galeão, Rio de Janeiro
- 19 target sites, spanning 11 km to ≈ 2,100 km

Nine of them inside the Rio basin; ten strung north along a corridor that ends in equatorial Maranhão. The network is dense in one place and empty almost everywhere else — which is the problem itself, drawn.

WHERE I WANT PUSHBACK · THE THREE OPEN QUESTIONS I BRING TO THIS SYMPOSIUM

- How I **individuate and operationalize** the similarity relations.
- The **epistemic status** I assign to a gate’s verdict.
- How to frame the **ontology–learning interface** so it speaks to both communities at once.

08 — RESEARCH VISION

Transferability is a structured, investigable phenomenon — and formal ontology has a concrete, testable role in explaining it.

Three commitments: ontology as a testable hypothesis, not assumed scaffolding · the **epistemology of similarity** — when inherited scientific classification carries inductive value, and when it must be re-grounded · neurosymbolic governance with an **accountable** symbolic layer.

REFERENCES

[1] Pan & Yang 2010, *IEEE TKDE* 22(10). [2] Weiss, Khoshgoftaar & Wang 2016, *J. Big Data* 3(1). [3] Guarino, Oberle & Staab 2009, *Handbook on Ontologies*. [4] Guizzardi 2005, PhD thesis, Univ. of Twente. [5] Tididi & Schlobach 2022, *Artificial Intelligence* 302. [6] Beck et al. 2018, *Scientific Data* 5. [7] Peel, Finlayson & McMahon 2007, *HESS* 11. [8] Whittaker 1975, *Communities and Ecosystems*. [9] Gelaro et al. 2017, *J. Climate* 30(14) (MERRA-2). Also cited in the paper: Garcez & Lamb 2023; Kautz 2022; Conradi et al. 2020; Fick & Hijmans 2017.

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